

Juhyun Bae Research Portfolio



Juhyun Bae
배주현

M.S. Candidate
Department of Electrical Engineering
Korea University

Deep Image Processing Lab

Research Interests

- Image Restoration
- Computational Photography
- Data Augmentation
- Deep Learning

Honors & Awards

- NRF Master's Student Research Grant, 2026
- 2nd Place, Samsung AI Challenge 2024
- Honorable Mention, Hanium ICT Mentoring Project Contest 2022

Research & Publications

- Trajectory-aware Differentiable Blur Augmentation for Image Deblurring (Target Venue: ICLR 2027, Ongoing)
- DnSAD: Defocus and Spatially Adaptive Deblurring for Display Mura Compensation, Journal of the Society for Information Display, 2026 (Co-first Author)
- Intrinsic Decomposition-based Low-Light Image Enhancement, IEEE Signal Processing Letters, 2025

Selected Projects

- Vision-based Safety Equipment Inspection (KIST AI-Robot Institute, 2023)
- Samsung AI Challenge, Machine Learning Force Fields (2024)
- Image Aesthetics Assessment using Vision-Language Models (2024)

Education

Korea University

M.S. Candidate in Electrical Engineering
(Advisor: Prof. Seung-Won Jung)
Feb, 2025 – Present

Seoul National University of Science and Technology

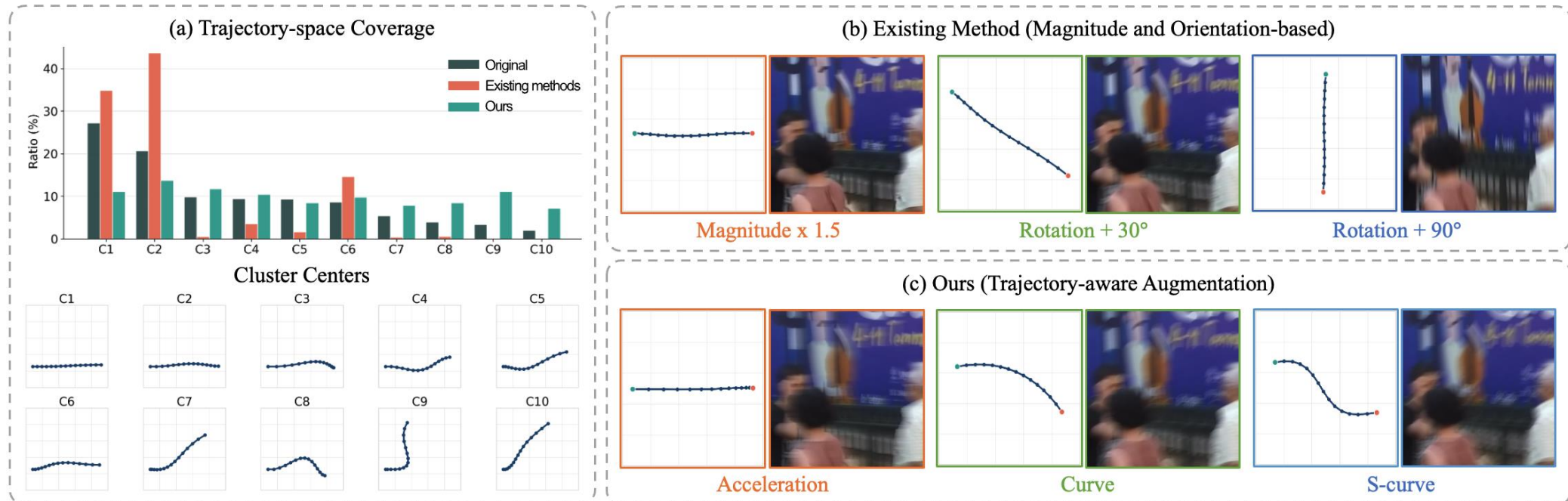
B.S. in Computer Science and Engineering
GPA: 4.37 / 4.50
Graduated as Valedictorian
Mar, 2020 - Feb, 2025

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

Target Venue: ICLR 2027

Problem & Motivation

- Motion blur is caused by camera shake and object motion during image exposure.
- Blur augmentation synthesizes additional blur from sharp images to expand deblurring training data.
- Existing blur augmentation varies only motion magnitude and orientation, limiting trajectory diversity.
- Trajectory-aware blur augmentation is proposed to enrich the motion trajectory space.



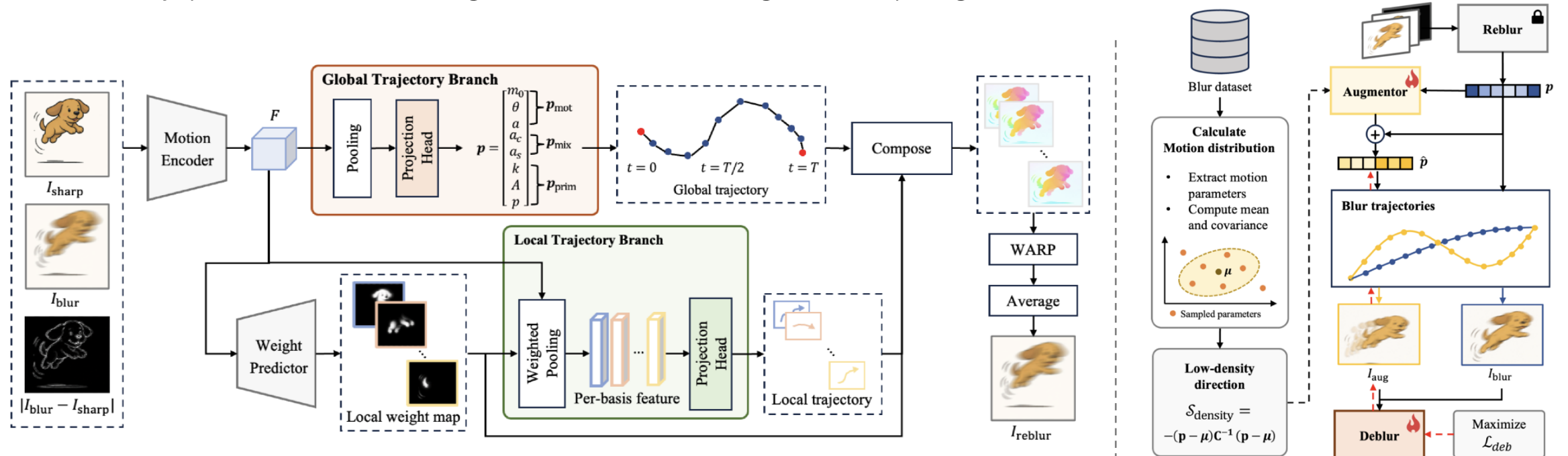
Comparison of trajectory diversity between existing blur augmentation and the proposed method.

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

Target Venue: ICLR 2027

Methods

- Train a trajectory-aware reblur network to model diverse motion trajectories, and utilize it for differentiable blur augmentation during deblurring training.
- Jointly model global camera motion and local object motion to generate realistic spatially varying motion blur.
- Jointly optimize learnable motion augmentation with the deblurring model to improve generalization.



Overview of the proposed trajectory-aware blur augmentation framework

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

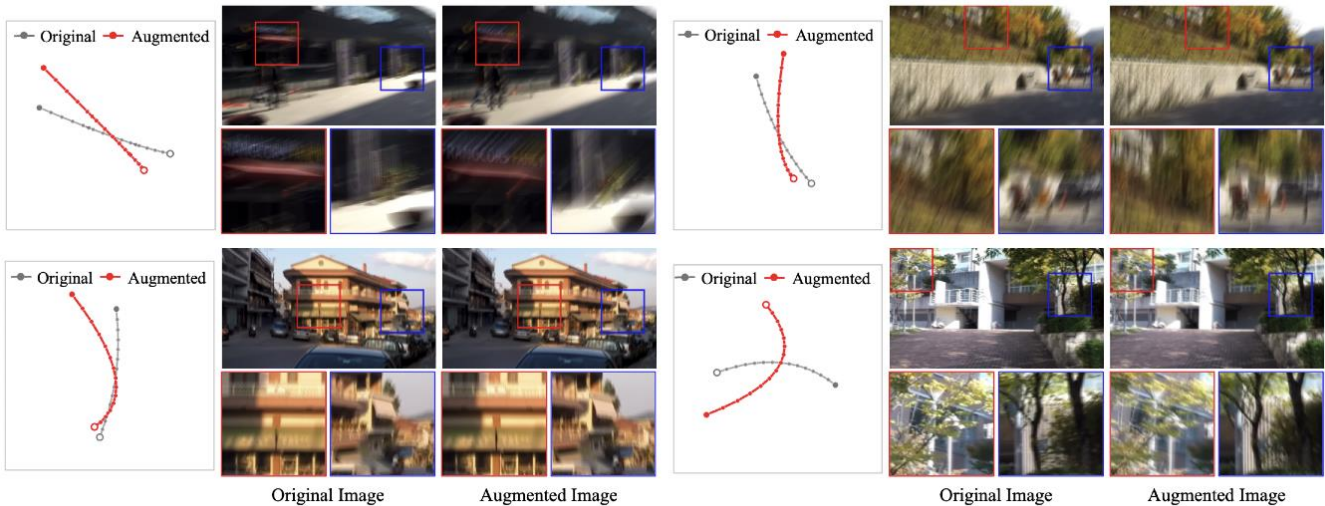
Target Venue: ICLR 2027

Results

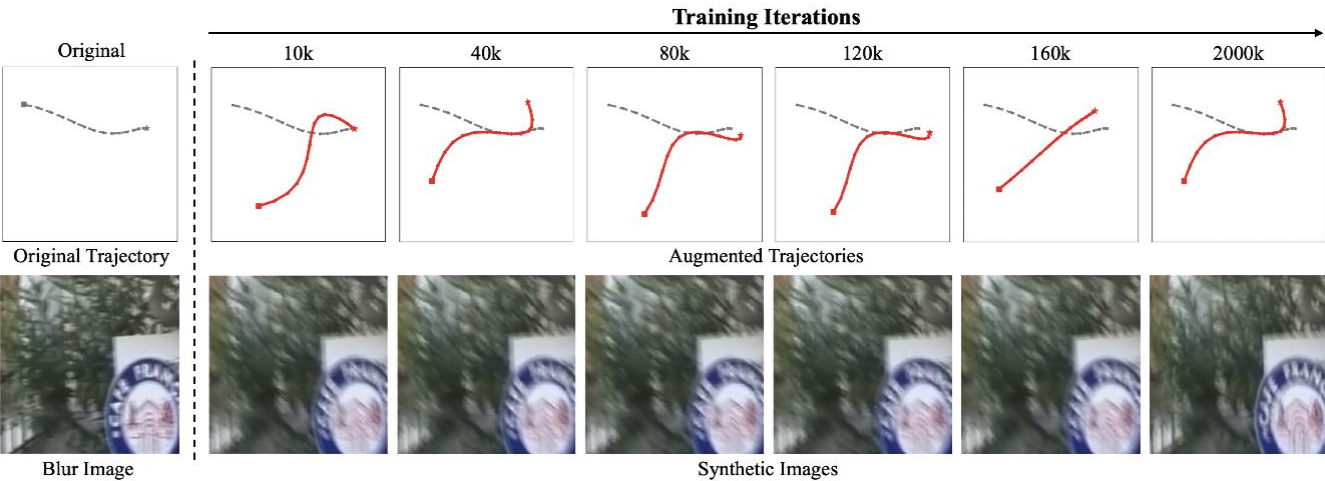
- +2.45 dB over baseline (GoPro 10%)
- Improved performance on HIDE benchmark

Model	GOPRO 10 %		GOPRO 100%	
	PSNR	SSIM	PSNR	SSIM
MIMO-UNet+	28.92	0.915	32.44	0.957
+ ID-Blau	30.61	0.940	32.93	0.961
+ Ours	31.37	0.947	33.10	0.962
HIDE (Test only)				
MIMO-UNet+	26.71	0.877	30.00	0.930
+ ID-Blau	28.50	0.910	30.68	0.938
+ Ours	28.98	0.919	30.71	0.938

Performance comparison of the proposed model and baseline methods.



Visualization of trajectory-aware augmentation and the corresponding blur variations.



Evolution of learnable trajectory augmentation during deblurring training.

Defocus and Spatially Adaptive Deblurring for Display Mura Compensation

Journal of the Society for Information Display, 2026 (Co-first author)

Project

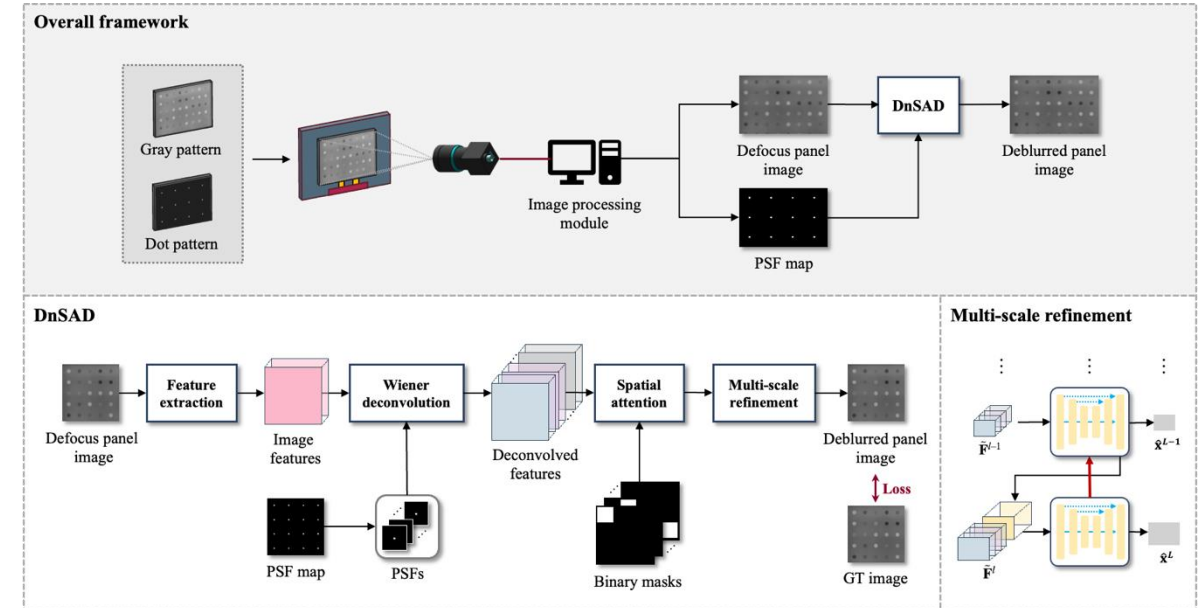
- Developed DnSAD as part of an LG Display industry-academia project to restore spatially varying defocus blur for display mura compensation.

My Contributions

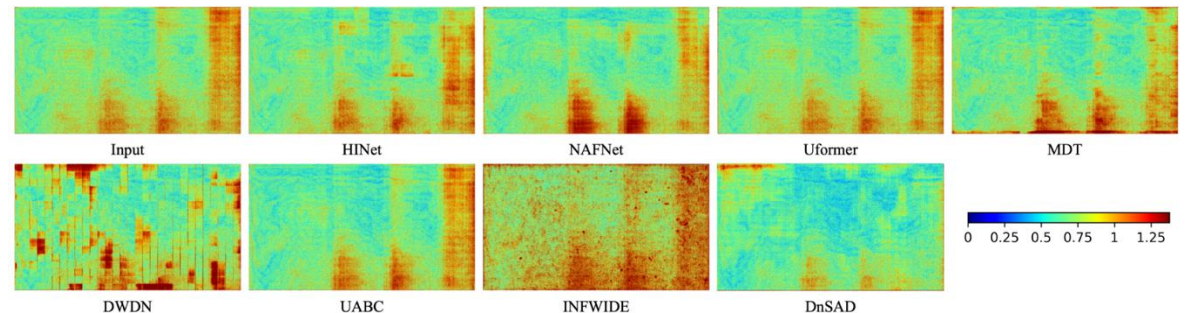
- Designed a PSF-aware deblurring framework leveraging measured PSF maps for spatially varying blur restoration.
- Introduced spatial attention and multi-scale refinement to effectively restore high-frequency mura patterns.
- Constructed a real-world OLED panel dataset and validated the framework for LG Display mura compensation.

Results

- Achieved 36.41 dB PSNR and 0.6474 MAE-MURA on real-world OLED panel datasets.
- Outperformed existing blind and non-blind deblurring methods.



Overview of the proposed PSF-aware deblurring framework (DnSAD).



Qualitative Results on Display Mura Compensation.

Defocus and Spatially Adaptive Deblurring for Display Mura Compensation

Journal of the Society for Information Display, 2026 (Co-author)

Project

- Developed ResCue, a residual-guided curriculum learning framework for low-light image enhancement.
- Utilized intrinsic decomposition residuals as difficulty cues to progressively train models from easy to hard samples.

My Contributions

- Contributed to designing the residual-based curriculum strategy and training pipeline.
- Conducted baseline training, comparative experiments, and ablation studies across multiple LLIE models.
- Analyzed experimental results and participated in discussions on model and training strategy design.

Results

- Achieved up to +1.27 dB PSNR improvement and published in IEEE Signal Processing Letters, 2025.



Overview of the proposed residual-guided curriculum learning framework (ResCue).

Methods	DICM	LIME	MEF	NPE	VV
HVI-CIDnet	4.13/31.49	4.20/18.43	3.93/21.28	3.83/21.09	3.37/29.39
+ ResCue	4.06/28.95	4.03/16.55	3.68/20.83	3.76/18.12	3.39/30.73

Quantitative results on unpaired datasets.